COM111 Tutorial 1b

1. Find i_1, i_2, and i_3 in Fig. 1

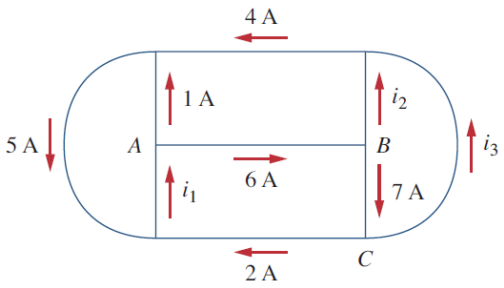


Fig. 1

2. Determine i_1 and i_2 in the circuit of Fig. 2.

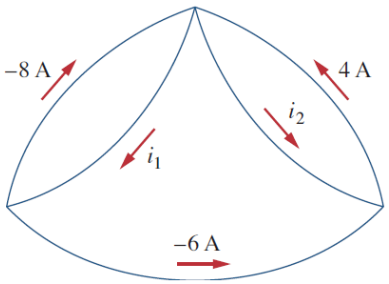


Fig. 2

3. In the circuit of Fig. 3, calculate V_1 and V_2.

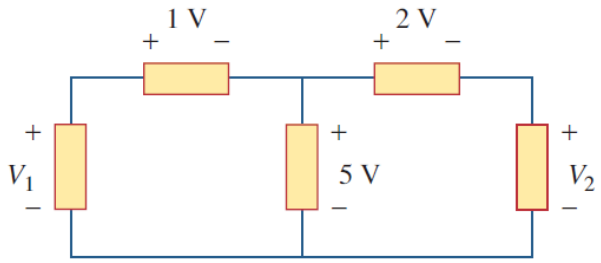


Fig. 3

4. In the circuit in Fig. 4, obtain v_1, v_2, and v_3.

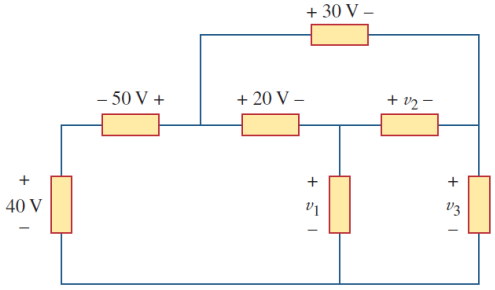


Fig. 4

5. For the circuit in Fig. 5, use KCL to find the branch currents I_1 to I_4.

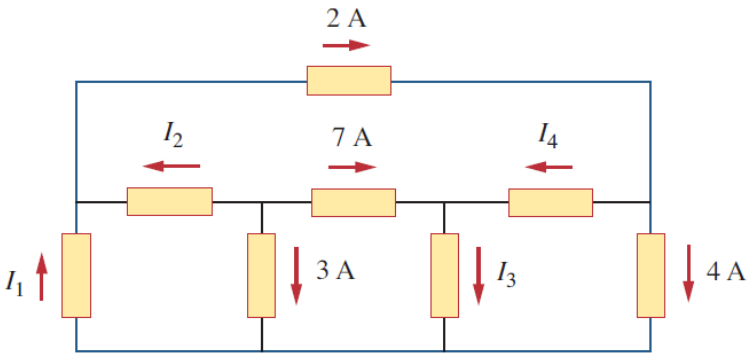


Fig. 5

6. Calculate *v* and *ix* in the circuit of Fig. 6

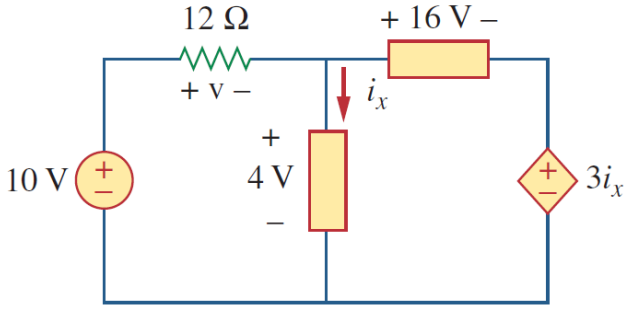


Fig. 6

7. Obtain *v*1, *v*2, and *v*3 in the circuit of Fig. 7.

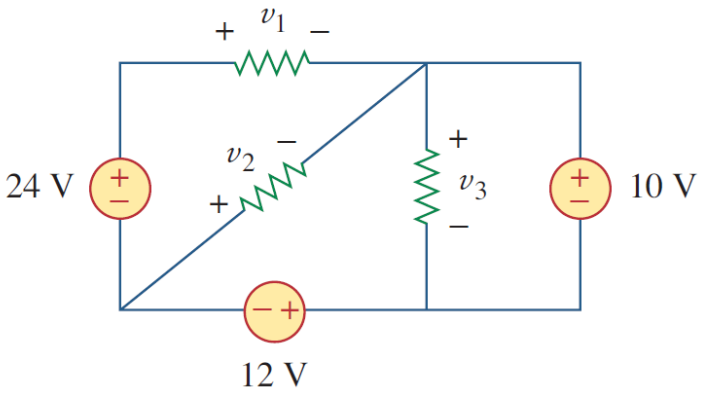


Fig. 7

8. From the circuit in Fig. 8, find I, the power dissipated by the resistor, and the power supplied by each source.

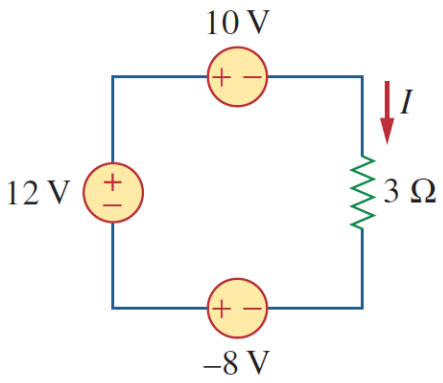


Fig. 8

9. For the circuit in Fig. 9, find Vo/Vs in terms of α, R1, R2, R3, and R4. If R1 = R2 = R3 = R4, what value of α will produce |Vo/Vs| = 10?

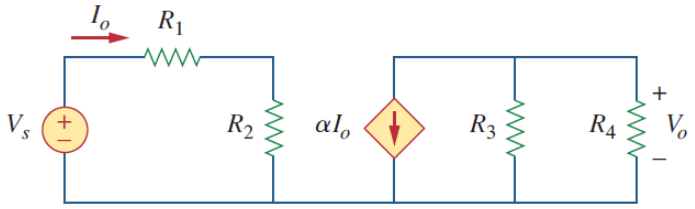


Fig. 9

10. For the circuit in Fig. 10, i_o = 3 A. Calculate i_x and the total power absorbed by the entire circuit.

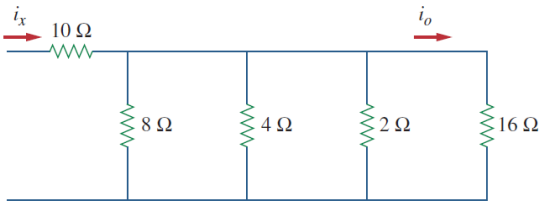


Fig. 10

11. Find *R*eq for the circuit in Fig. 11.

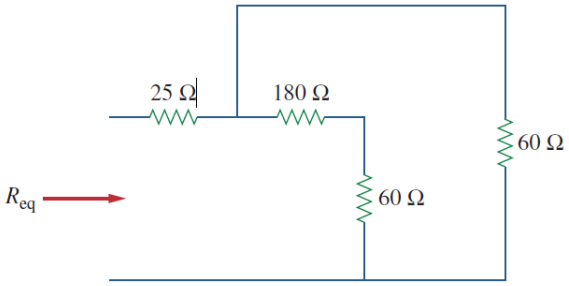


Fig. 11

12. For the circuit in Fig. 12, determine *i*1 to *i*5.

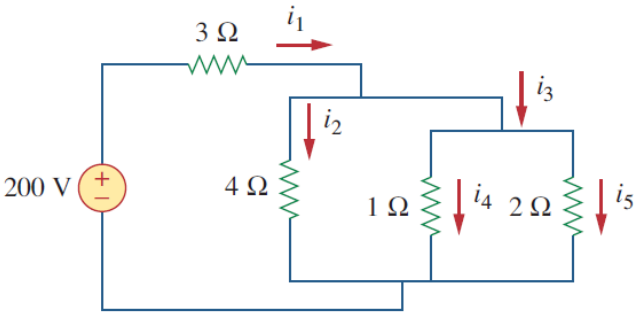


Fig. 12

13. Find *i*1, *i*2, *i*3, and *i*4 in the circuit in Fig. 13

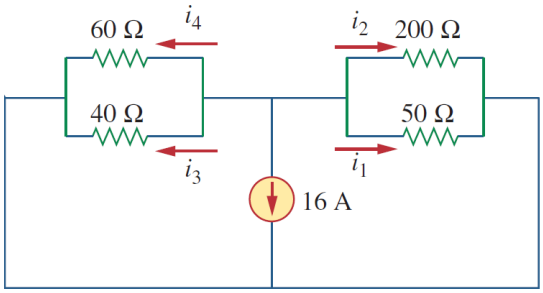


Fig. 13

14. Calculate *Vo* and *Io* in the circuit of Fig. 14.

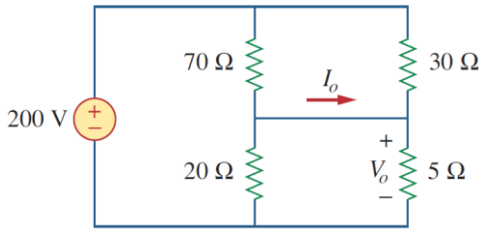


Fig. 14

15. If *R*eq = 50 Ω in the circuit of Fig. 15, find *R*.

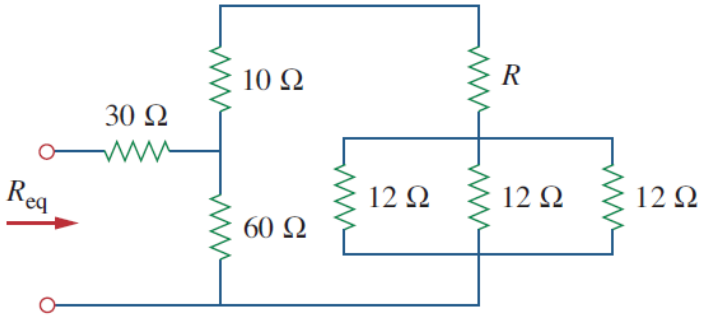


Fig. 15

16. Find the equivalent resistance *R*ab in the circuit of Fig. 16.

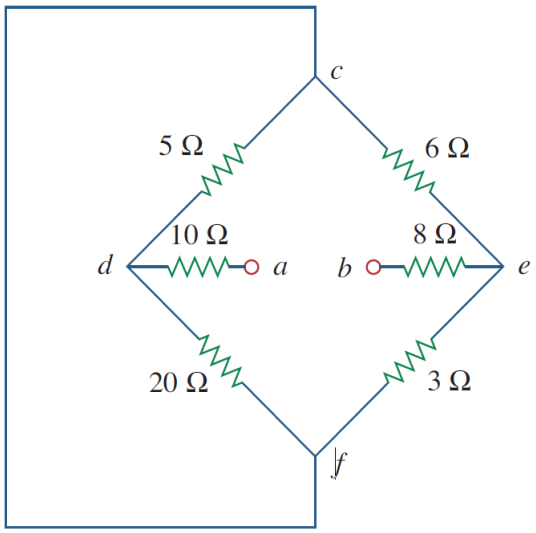


Fig. 16